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CMI
BVR

Analysis I/B1
Quiz 1

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12:30 - 13:10

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If you are using a theorem, quote it precisely.

1. Suppose (x_n) is a Cauchy sequence. Define $y_n = x_{5n}$ for $n \geq 1$. Assume $y_n \rightarrow a$. Show $x_n \rightarrow a$. [3]
(Thus $y_1 = x_5, y_2 = x_{10}, y_3 = x_{15}, \dots$).
2. Assume $x_n \rightarrow a$ and π is a permutation of $N = \{1, 2, \dots\}$, this means π is a bijection of N . Put $y_n = x_{\pi(n)}$. Show that $y_n \rightarrow a$. [3]
3. Explain true or false and justify your answer. [2+2]

(a) $n^{1000} \frac{1}{(1000)^n} \rightarrow 1,$

(b) If $x_n \geq y_n$ for infinitely many n , then $\limsup x_n \geq \limsup y_n$.